A COMPUTATION-FREE PROOF OF THE CHOW RANK OF THE FIVE-BY-FIVE PERMANENT

RONGYU XU

Abstract. We prove that the ordinary Chow rank of the permanent of a generic five-by-five matrix is sixteen over every field of characteristic zero. The proof combines a rectangular image-span inequality with a local polarization estimate for quintic products. A closed count of squarefree triples, a uniform monomial-divisor bound, and explicit derivatives in four variables establish the local estimate. No computer-generated rank, finite search, or stored certificate is a premise. In particular, the four normal-form catalecticant ranks need not be determined separately. The argument does not assert an equality for border Chow rank.

1. Statement and specialization conventions

Let $\mathbb{K}$ initially be algebraically closed of characteristic zero. Write $[m] = \{1, \dots, m\}$, and set

$$V = \operatorname{span}_{\mathbb{K}}\{x_{ij} : i, j \in [5]\}, \qquad f = \operatorname{perm}_5 = \sum_{\sigma \in S_5} \prod_{i=1}^{5} x_{i,\sigma(i)}.$$

Thus $\dim V = 25$, and elements of V are linear forms on V^*. The ordinary Chow rank, also called product rank, of a degree-m form is the least number of nonzero products of m linear forms summing to it; scalar coefficients may be absorbed into a factor.

Theorem 1.1. *For every field $\mathbb{K}$ of characteristic zero,*

$$\operatorname{ChowRank}_{\mathbb{K}}(\operatorname{perm}_5) = 16.$$

The factors in a Chow term may involve all twenty-five variables. This is distinct from the tensor rank of the permanent in five separate row spaces. Han and Song prove that the latter rank is sixteen [1]; that lower bound is not used here. In fact, the component with degree one in each row of the single Chow term $\prod_{j=1}^{5}(\sum_{i=1}^{5} x_{ij})$ is the permanent itself. For background on product rank, see [3].

For a form $h \in \operatorname{Sym}^m V$ let

$$C_{a,m-a}(h) : \operatorname{Sym}^a V^* \longrightarrow \operatorname{Sym}^{m-a} V, \quad D \longmapsto D(h),$$
$$\mathcal{D}_j(h) = \operatorname{im} C_{m-j,j}(h).$$

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All derivative spaces below are *actual* images of these maps, not formal spans of factor products. For a space of forms G, ∂G denotes the span of all first directional derivatives. The differential pairing between symmetric powers is perfect in characteristic zero. Under it $C_{a,m-a}(h)^* = C_{m-a,a}(h)$, since $D'D(h) = DD'(h)$. In particular these maps have the same rank.

We record precisely how specialization is used. For a polynomial matrix, rank at a special parameter value is no larger than rank for general values: every minor that vanishes identically still vanishes at the special value. Subspaces of a fixed dimension have limits under a diagonal one-parameter action. One can see existence by taking the least-weight nonzero term in their Plücker coordinates, after removing a common power of the parameter. The limit is fixed by the action. When the weights on the underlying basis vectors are distinct, every fixed subspace is a span of basis vectors, so the limit is coordinate. The same argument applies to several subspaces simultaneously. Locally on a Grassmannian, bases of the subspaces and their quotients give regular matrices; the minor criterion therefore applies to maps between these quotients as well. Thus an intersection or a kernel dimension can increase, but cannot decrease, at a special point. Only these elementary facts about limits are required below; no classification of algebraic-group orbits is used.

2. The permanent's derivative spaces

Put $E_j = \mathcal{D}_j(f)$.

Lemma 2.1. *One has*

$$\dim E_2 = \dim E_3 = 100, \qquad \partial E_2 = V,$$

$$E_2^{(1)} := \{g \in \operatorname{Sym}^3 V : \partial g \subset E_2\} = E_3.$$

Every nonzero quadratic in E_2 has matrix rank at least four. For any $L \subset V$ of dimension at most five,

$$\dim(E_2 \cap \operatorname{Sym}^2 L) \le 1. \tag{2.1}$$

Proof. The quadratic and cubic subpermanents form bases of E_2 and E_3. Different row-set/column-set pairs have disjoint matching-monomial supports, and their numbers are $\binom{5}{2}^2 = \binom{5}{3}^2 = 100$. In particular,

$$E_2 = \operatorname{span}\{x_{ia}x_{jb} + x_{ib}x_{ja} : i < j,\ a < b\}.$$

Differentiating a rectangle at one corner gives its opposite variable, so $\partial E_2 = V$.

Suppose every first derivative of a cubic g belongs to E_2. A cubic monomial with two factors in one row or column, including a repeated factor, is impossible: differentiation by the remaining factor produces a quadratic monomial forbidden in E_2. For a fixed partial derivative and a fixed resulting monomial there is a unique source monomial, so its coefficient cannot be cancelled by a different source. Its derivative multiplicity is nonzero. Every surviving monomial is consequently a three-edge matching. For fixed three

rows and three columns, swapping the assignments of two rows leaves the third factor unchanged. Differentiating by that common factor forces the two matching coefficients to agree. Transpositions connect all six permutations, so their coefficients all agree. This proves $E_2^{(1)} = E_3$, including the immediate reverse inclusion.

For the remaining assertions use the row-column scaling

$$\tau_t(x_{ij}) = t^{r_i+c_j} x_{ij}, \qquad r_i = 2^{i-1}, \quad c_j = 2^{5+j-1}, \quad t \neq 0. \tag{2.2}$$

The twenty-five variable weights are distinct. Both monomials of each rectangle acquire the same weight, and the one hundred rectangle weights are distinct, by uniqueness of binary expansion. Thus τ_t preserves E_2, and any nonzero $q \in E_2$ has a projectively normalized limit equal to a nonzero rectangle. Invertible substitutions and nonzero scalar multiplication preserve quadratic rank; specialization cannot increase it. A rectangle is the sum of two products on four independent variables, with a nonsingular four-dimensional symmetric matrix. Its rank is four, proving the minimum-rank claim.

For $a = \dim L \leq 5$, take the coordinate limit L_0 of $\tau_t L$ in $\mathrm{Gr}(a, V)$. The intersection in (2.1) is the kernel of $\mathrm{Sym}^2 L \to \mathrm{Sym}^2 V/E_2$. Because E_2 is preserved and kernel dimension cannot decrease at the limit, it suffices to bound the intersection for L_0. Let $\Omega \subset [5] \times [5]$ be its set of a variables. Distinct rectangles have disjoint quadratic monomial supports, so $E_2 \cap \mathrm{Sym}^2 L_0$ has as a basis precisely the rectangles whose four corners lie in Ω. Two different rectangles share at most two corners: three corners determine the fourth. They therefore require at least six variables. Since $|\Omega| \leq 5$, at most one rectangle occurs. □

3. Local polarization of a quintic product

Let $T = \ell_1 \cdots \ell_5 \neq 0$, and put

$$L = \mathrm{span}(\ell_1, \ldots, \ell_5), \quad \ell = \dim L,$$
$$F = \mathcal{D}_2(T), \quad U = \mathcal{D}_3(T), \quad u = \dim F = \dim U \leq 10.$$

The equality uses the differential pairing. The upper bound follows because third derivatives are linear combinations of the ten pair products of the five factors. Also $F \subset \mathrm{Sym}^2 L$ and $\partial U \subset F$. Let $R = F \cap E_2$; by Lemma 2.1, $\dim R \leq 1$, and every nonzero element of R has quadratic rank at least four.

For a surjection $P : L \to D$, set $d = \dim D$, $S = \ker P$, and $k = \dim S = \ell - d$. Define

$$\delta : U \to L \otimes F, \qquad \delta(g) = \sum_j y_j \otimes \partial_{y_j^*} g, \qquad \beta = (P \otimes q_R)\delta,$$

where $q_R : F \to F/R$ is the quotient map. The displayed polarization is independent of the choice of basis (y_j) of L. It is injective, since multiplication of its two factors sends $\delta(g)$ to $3g$.

Lemma 3.1 (A small complementary span)**.** *If $k \le 2$, then*

$$\ker \beta = U \cap \operatorname{Sym}^3 S, \qquad \operatorname{rank} \beta \ge u - \binom{k+2}{3}.$$

Proof. Choose a basis of S, extended by complementary variables $z_1, \dots, z_d$ in L. For a kernel cubic the derivatives $\partial_{z_j^*} g$ all lie in R. If any is nonzero, a change of complementary coordinates makes these derivatives $q, 0, \dots, 0$ for some $0 \neq q \in R$, since $\dim R \le 1$. Vanishing of the last $d-1$ derivatives implies that g, and then q, uses only S and z_1. Hence $\operatorname{rank} q \le k+1 \le 3$, a contradiction. All complementary derivatives must vanish, which is exactly $g \in \operatorname{Sym}^3 S$. The converse is immediate, and $\dim \operatorname{Sym}^3 S = \binom{k+2}{3}$. □

Lemma 3.2 (A uniform divisor bound)**.** *If $1 \le \ell \le 4$, then*

$$u \ge \binom{\ell}{2} + 1. \tag{3.1}$$

Proof. Choose ℓ independent factors as coordinates $y_1, \dots, y_\ell$. Scale the coordinates with distinct weights and divide each factor by its own least power of the parameter. Each normalized factor tends to a nonzero coordinate multiple. For nonzero parameter their product is a scalar multiple of an invertible transform of T. The limit is a nonzero monomial $c \prod_i y_i^{\alpha_i}$, with $\alpha_i \ge 1$ and $\sum_i \alpha_i = 5$; the initially chosen coordinate factors ensure that none of the ℓ variables is lost. Its nonzero second partials are indexed by its distinct degree-two divisors and yield distinct cubic monomials. The number of these divisors is

$$\binom{\ell}{2} + \#\{i : \alpha_i \ge 2\}.$$

Since $5 > \ell$, at least one exponent is at least two. Rank of the catalecticant cannot increase under specialization, giving (3.1). No enumeration of positive partitions is involved. □

Lemma 3.3 (Five independent factors)**.** *If $\ell = 5$, then $u = 10$ and $\operatorname{rank} \beta \ge 2d$.*

Proof. Use the five factors as coordinates $z_1, \dots, z_5$. The spaces U and F are respectively the squarefree cubic and quadratic spaces. We prove the estimate for every subspace $R \subset F$ of dimension at most one, without its minimum-rank condition.

Replace P by the canonical quotient $L \to L/S$, which does not change the rank. Simultaneously specialize S and R by the diagonal action with variable weights 2^{i-1}. The weights on variables and on squarefree quadratic monomials are distinct. Both limits are coordinate subspaces, and U, F are preserved throughout. By polarization equivariance the rank is constant for $t \neq 0$, and it cannot increase at the limit. For clarity, the kernel of the induced quotient map on $L \otimes F$ is $S \otimes F + L \otimes R$, of constant dimension $10k + 5 \dim R - k \dim R$. The specialization comparison is therefore

between fixed-rank subspace quotients, not between arbitrarily chosen bases of changing dimension. The limit of R need not retain quadratic rank four; our stronger statement does not require that.

Write $z_J = \prod_{j\in J} z_j$. At the limit let $B \subset [5]$ index the basis of S, and set $b = |B| = 5 - d$. If $R \neq 0$, let $e \in \binom{[5]}{2}$ be the edge with $R = \operatorname{span}(z_e)$. A source triple $I \in \binom{[5]}{3}$ has components $z_i \otimes z_{I\setminus\{i\}}$, for $i \in I$. A component survives exactly when $i \notin B$ and its complementary edge is not e. Different triples have disjoint component supports, so the rank is the number of surviving columns. A column is zero if $I \subset B$, or if it has exactly one vertex outside B and its complementary edge is $e \subset B$. A triple with two or more vertices outside B cannot be killed by a single edge. Consequently the coordinate rank is exactly

$$10 - \binom{b}{3} - d\,\mathbf{1}_{\{R\neq 0,\ e\subset B\}}. \tag{3.2}$$

If $b = 0, 1$, the indicator and binomial coefficient vanish, so this rank is ten, at least $2d$. If $2 \le b \le 5$, subtracting $2d = 2(5 - b)$ from its lower bound gives

$$10 - \binom{b}{3} - (5 - b) - 2(5 - b) = \frac{(5 - b)(b^2 + 2b - 6)}{6} \ge 0.$$

This proves the assertion without a coordinate-case table. □

Lemma 3.4 (One new direction in a four-dimensional factor span)**.** *If $\ell = 4$ and $d = 1$, then*

$$\operatorname{rank}\beta + \frac{10 - u}{2} \ge 2.$$

Proof. Choose four independent factors as coordinates. Up to a nonzero scalar, rescaling and permuting them makes the product

$$T_s = x_1x_2x_3x_4(x_1 + \cdots + x_s), \qquad 1 \le s \le 4.$$

Identify the one-dimensional target of P with $\mathbb{K}$. Then β is differentiation in a nonzero direction $\lambda = \sum_{j=1}^{4} a_j\partial_{x_j}$, followed by the quotient by R. Write $H_{ij} = \partial_{x_i}\partial_{x_j}T_s$.

For $s = 1$, $H_{22} = H_{33} = H_{44} = 0$, so $u \le 7$. For $s = 2$, one has

$$H_{33} = H_{44} = 0, \quad H_{11} = 2x_2x_3x_4, \quad H_{22} = 2x_1x_3x_4, \quad H_{12} = H_{11} + H_{22},$$

so again $u \le 7$. In both cases U contains $g_1 = x_1x_3x_4$ and $g_2 = x_2x_3x_4$: for $s = 1$ use $H_{12}/2, H_{11}/2$, and for $s = 2$ use $H_{22}/2, H_{11}/2$. Their supports together contain all four variables. Directly, $\partial_\lambda g_1 = \partial_\lambda g_2 = 0$ forces all four a_j to vanish, by comparing their distinct pair monomials. Thus at least one of these derivatives is nonzero. It uses at most three variables and has quadratic rank at most three, so it cannot lie in R. It follows that $\operatorname{rank}\beta \ge 1$, and $\operatorname{rank}\beta + (10 - u)/2 \ge 1 + 3/2 > 2$.

For $s = 3$, the identities $H_{jj} = 2\prod_{i\neq j} x_i$, for $j = 1, 2, 3$, show that U contains $x_1x_2x_4, x_1x_3x_4, x_2x_3x_4$. If $a_4 \neq 0$, their derivatives modulo x_4L are

nonzero multiples of three distinct pair monomials. Their derivative span has dimension at least three, so its quotient by the at-most-one-dimensional R has dimension at least two. If $a_4 = 0$, relabel x_1, x_2, x_3 so that $a_1 \neq 0$. The derivatives of $x_1x_2x_4, x_1x_3x_4$ are

$$x_4(a_1x_2 + a_2x_1), \qquad x_4(a_1x_3 + a_3x_1).$$

They are independent. Every quadratic in their span is divisible by x_4, hence has rank at most two, so the span meets R trivially. Again $\operatorname{rank}\beta \geq 2$.

For $s = 4$, the same diagonal derivative identities show that U contains all four squarefree cubics. Choose $a_j \neq 0$. Differentiating the three cubics containing x_j, and reducing modulo x_jL, gives three independent pair monomials on the other three variables. Quotienting their derivative span by R leaves dimension at least two. For $s = 3, 4$, $u \leq 10$ now proves the asserted inequality. In particular, neither of their exact catalecticant ranks was needed. □

Proposition 3.5 (The local half-defect estimate)**.** *For every nonzero quintic Chow term and every quotient P of its actual factor span,*

$$\boxed{\operatorname{rank}\beta + \frac{10-u}{2} \geq 2d.} \tag{3.3}$$

Proof. For $d = 0$ this is immediate, since $u \leq 10$. If $\ell = 5$, use Lemma 3.3. Suppose $\ell \leq 4$ and $d > 0$. If $k = \ell - d \leq 2$, Lemmas 3.1 and 3.2 give

$$\begin{aligned}\operatorname{rank}\beta + \frac{10-u}{2} - 2d &\geq 5 + \frac{u}{2} - \binom{k+2}{3} - 2(\ell - k) \\ &\geq \frac{(\ell-4)(\ell-5)+2}{4} + 2k - \binom{k+2}{3} \geq 0.\end{aligned}$$

Indeed $(\ell-4)(\ell-5) \geq 0$ for $1 \leq \ell \leq 4$, and $2k - \binom{k+2}{3}$ is respectively $0, 1, 0$ for $k = 0, 1, 2$. The only remaining possibility is $\ell = 4, d = 1, k = 3$, covered by Lemma 3.4. Thus no integer-interval or partition table is required to complete the cases. □

4. The global dimension contradiction

Lemma 4.1 (Rectangular image spans)**.** *For linear maps $A_i : X^* \to Y$ of ranks u_i, put $D_Y = \dim \sum_i \operatorname{im} A_i$ and $D_X = \dim \sum_i \operatorname{im} A_i^*$. Then*

$$\operatorname{rank}\Big(\sum_i A_i\Big) \geq D_Y + D_X - \sum_i u_i.$$

Proof. Factor $A_i = B_iC_i$ through a vector space of dimension u_i, with B_i injective and C_i surjective. The concatenated maps $B : \bigoplus_i \mathbb{K}^{u_i} \to Y$ and $C : X^* \to \bigoplus_i \mathbb{K}^{u_i}$ have ranks D_Y, D_X, respectively, and $BC = \sum_i A_i$. Here $\operatorname{im} A_i^* = \operatorname{im} C_i^*$, since B_i^* is onto, justifying the rank of C. Since $\dim(\operatorname{im} C \cap \ker B) \leq \dim \ker B = \sum_i u_i - D_Y$, the rank of BC is at least $D_X + D_Y - \sum_i u_i$. □

Suppose an actual Chow decomposition $f = \sum_{i=1}^{N} T_i$ is given. Let

$$F_i = \mathcal{D}_2(T_i), \quad U_i = \mathcal{D}_3(T_i), \quad u_i = \dim F_i = \dim U_i, \quad \Delta = \sum_i (10 - u_i) \geq 0,$$

$$F = \sum_i F_i, \quad U = \sum_i U_i, \qquad \dim F = 100 + h_2, \quad \dim U = 100 + h_3.$$

Linearity of catalecticants gives $E_2 \subset F$, $E_3 \subset U$, so $h_2, h_3 \geq 0$. Apply Lemma 4.1 to $A_i = C_{3,2}(T_i)$. Their adjoints have images U_i, and their sum $C_{3,2}(f)$ has rank one hundred. Hence

$$h_2 + h_3 \leq 10N - 100 - \Delta. \tag{4.1}$$

Let $q : F \to F/E_2$ and use polarization to define

$$b : U \longrightarrow V \otimes (F/E_2), \qquad b(g) = (1_V \otimes q)\delta(g).$$

Since $\partial U \subset F$, this map is well defined. Its kernel is $U \cap E_2^{(1)} = E_3$, by Lemma 2.1 and $E_3 \subset U$. Therefore

$$\operatorname{rank} b = h_3. \tag{4.2}$$

Order the terms arbitrarily. Put $L_i = \operatorname{span}\{\text{factors of } T_i\}$, $W_i = L_1 + \cdots + L_i$, $W_0 = 0$, and $d_i = \dim(W_i/W_{i-1})$. We have $W_N = V$, because $V = \partial E_2 \subset \partial F \subset \sum_i L_i$. Thus $\sum_i d_i = 25$. Set $Z_i = b(U_i)$ and $H_i = Z_1 + \cdots + Z_i$. As $Z_j \subset L_j \otimes (F/E_2)$, projection to $(W_i/W_{i-1}) \otimes (F/E_2)$ kills H_{i-1}. It follows that

$$\dim H_i - \dim H_{i-1} \geq \operatorname{rank}\big((P_i \otimes q|_{F_i})\delta|_{U_i}\big),$$

where $P_i : L_i \to L_i/(L_i \cap W_{i-1})$ has rank d_i. The natural map $F_i/(F_i \cap E_2) \hookrightarrow F/E_2$ is injective. Thus the rank on the right is exactly the local symbol rank in Proposition 3.5, not an assumed independence of the unprojected images Z_i. Since $H_N = b(U)$, summation yields

$$h_3 \geq \sum_i \left(2d_i - \frac{10 - u_i}{2}\right) = 50 - \frac{\Delta}{2}. \tag{4.3}$$

Combining (4.1) and (4.3), we obtain

$$\boxed{h_2 + \frac{\Delta}{2} \leq 10N - 150.} \tag{4.4}$$

For $N < 15$ the right side is negative, impossible. If $N = 15$, nonnegativity forces $h_2 = \Delta = 0$. Then $F = E_2$, and every $u_i = 10$, because each $10 - u_i$ is nonnegative. Consequently

$$F_i \subset E_2 \cap \operatorname{Sym}^2 L_i, \qquad \dim F_i = 10,$$

contradicting (2.1), since $\dim L_i \leq 5$. We conclude that $N \geq 16$.

5. The upper bound and the logical boundary

Glynn's identity [2] specializes to

$$\operatorname{perm}_5 = \frac{1}{16} \sum_{\substack{\epsilon \in \{\pm 1\}^5 \\ \epsilon_1 = 1}} \left(\prod_{r=1}^{5} \epsilon_r \right) \prod_{c=1}^{5} \left(\sum_{r=1}^{5} \epsilon_r x_{rc} \right). \tag{5.1}$$

Here is a direct coefficient proof. Every expansion monomial chooses a row in each of the five columns. Let m_r count its row-r choices. The coefficient of this monomial is

$$\frac{1}{16} \prod_{r=2}^{5} \sum_{\epsilon_r = \pm 1} \epsilon_r^{m_r+1}.$$

It vanishes unless m_2, m_3, m_4, m_5 are all odd. They are nonnegative integers whose total, together with m_1, is five. The four odd counts must each be one, since making one at least three would require total at least six. Then $m_1 = 1$. Precisely the permutation monomials survive, each with coefficient one. Equation (5.1) has sixteen Chow terms and proves the upper bound. Together with the preceding section it proves Theorem 1.1 over algebraically closed characteristic-zero fields. For any characteristic-zero field, scalar extension preserves a hypothetical shorter decomposition, giving the same lower bound over that field; the rational upper bound already holds there.
Scope and disclosure. Every estimate used above is proved in the text. The triple formula (3.2) replaces the coordinate scan, Lemma 3.2 replaces partitions, and Lemma 3.4 uses explicit derivatives rather than four exact catalecticant ranks. Optional programs are regression checks, not proof premises. No border-rank equality or general-size formula is claimed.

This AI-assisted revision develops the author's October 16, 2026 short proof in `permanent-chow-rank-small-n`, `submissions/20261016-r2/perm5/main.tex`; that frozen release is unchanged. ChatGPT assisted argument development, internal auditing, exposition, and verification materials. Internal review is not external peer review or formal proof-assistant certification. Responsibility for the final manuscript rests with the author.

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